(vii) In the expansion of $(a+b)^n, n \in \mathbb{N}$, the greatest coeff cient is ${}^nC_{\frac{n}{2}}$ if n is even and the greatest coeff cients are ${}^nC_{\frac{n-1}{2}}$ or ${}^nC_{\frac{n+1}{2}}$, if n is odd.

(viii) In the expansion of $(a+b)^n, n \in \mathbb{N}$,
if n is even, the middle term is $T_{\frac{n}{2}+1} = {}^nC_{\frac{n}{2}}\, a^{n-\frac{n}{2}} b^{\frac{n}{2}}$. If n is odd, then the two middle terms are $T_{\frac{n-1}{2}+1}$ and $T_{\frac{n+1}{2}+1}$.

5.3 Particular cases of Binomial Theorem

(i) Replacing b by $(-b)$, in the binomial expansion of $(a+b)^n, n \in \mathbb{N}$, we get

$$\begin{aligned}(a-b)^n &= {}^nC_0\, a^n b^0 - {}^nC_1\, a^{n-1}b^1 + {}^nC_2\, a^{n-2}b^2 - \cdots \\ &\quad + (-1)^r\, {}^nC_r\, a^{n-r}b^r + \cdots + (-1)^n\, {}^nC_n\, a^0 b^n.\end{aligned}$$

Observe that the sign '+' and '−' appear alternately in the binomial expansion of $(a-b)^n$.

(ii) Replacing a by 1 and b by x, in the binomial expansion of $(a+b)^n, n \in \mathbb{N}$, we get

$$(1+x)^n = {}^nC_0 + {}^nC_1\, x + {}^nC_2\, x^2 + \cdots + {}^nC_r\, x^r + \cdots + {}^nC_n\, x^n.$$

In particular, when $x = 1$, ${}^nC_0 + {}^nC_1 + {}^nC_2 + \cdots + {}^nC_n = 2^n$.

If X is a set containing n elements, then we know that ${}^nC_r$ is the number of subsets of X having exactly r elements. So by adding ${}^nC_r$ for $r = 0, 1, 2, \ldots, n$ we get the number of subsets of X. So by using the above identity we see that **a set of n elements has 2^n subsets.**

(iii) $(1-x)^n = {}^nC_0 - {}^nC_1\, x + {}^nC_2\, x^2 - \cdots + (-1)^r\, {}^nC_r\, x^r + \cdots + (-1)^n x^n$. In particular, when $x = 1$, we get, ${}^nC_0 + {}^nC_2 + {}^nC_4 + \cdots = {}^nC_1 + {}^nC_3 + {}^nC_5 + \cdots = 2^{n-1}$

Example 5.1 Find the expansion of $(2x+3)^5$.

Solution:
By taking $a = 2x$, $b = 3$ and $n = 5$ in the binomial expansion of $(a+b)^n$ we get

$$\begin{aligned}(2x+3)^5 &= (2x)^5 + 5(2x)^4 3 + 10(2x)^3 3^2 + 10(2x)^2 3^3 + 5(2x)3^4 + 3^5 \\ &= 32x^5 + 240x^4 + 720x^3 + 1080x^2 + 810x + 243.\end{aligned}$$

Example 5.2 Evaluate 98^4.

Solution:
By taking $a = 100$, $b = 2$ and $n = 4$ in the binomial expansion of $(a-b)^n$ we get

$$\begin{aligned}98^4 &= (100-2)^4 \\ &= {}^4C_0\, 100^4 - {}^4C_1\, 100^3 2 + {}^4C_2\, 100^2 2^2 - {}^4C_3\, 100^1 2^3 + {}^4C_4\, 100^0 2^4 \\ &= 100000000 - 8000000 + 240000 - 3200 + 16 \\ &= 92236816.\end{aligned}$$